

Two Paths: Local to Cloud LLM APIs

Local models need no API key, data never leaves your machine, code switches seamlessly [Ollama, 2025] [Kwon et al., 2023]

Local Deployment (Ollama)

Zero Cost · Data Private · No Rate Limit

❑ ollama pull qwen3.5:2b

Download model ≈2.7GB (Q4 quantization)

❑ ollama serve

Start HTTP service localhost:11434

❑ Python Call (OpenAI format)

```
client = OpenAI(base_url="http://localhost:11434/v1")
model="qwen3.5:2b"
```

Hardware Reference

qwen3.5:0.8b ❑ 1.0GB (integrated GPU ok)

qwen3.5:2b ❑ 2.7GB (integrated GPU ok)

qwen3.5:9b ❑ 6.6GB (RTX 4060 8GB)

qwen3.5:27b ❑ 17GB (RTX 4090 24GB)

Inference Speed (RTX 4090):

2B: ~80-120 tok/s | 9B: ~50-80 tok/s

Cloud API

Pay per token · Better results · Rate limited

Major API Provider Comparison

Model	Input/1M	Output/1M
GPT-4o	\$2.50	\$10.00
GPT-4o mini	\$0.15	\$0.60
Claude 3.5 Sonnet	\$3.00	\$15.00
DeepSeek V3	¥1	¥2
Qwen-Max	¥2	¥6

Source: Official model pricing, Jan 2025

Cost Warning

100 req/min × 500in + 200out tok

GPT-4o: \$468/day ≈ \$14,000/month

Local 2B model: \$0/month

Seamless Switch

Just change 2 lines of code, everything else stays the same

```
Local: base_url="http://localhost:11434/v1"
api_key="ollama"
```

```
Cloud: base_url="https://api.openai.com/v1"
api_key="sk-..."
```